

1. A coupled partial melting Stokes-Darcy system. Strong form

$$\begin{aligned}
 (1.1) \quad & \tilde{\mathbf{v}}_r + \frac{k_0 \phi_f^{1+\Theta}}{\mu_f} \nabla (\phi_f^{-1/2} \tilde{q}_f) = 0 \\
 & \mu_s \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0 \\
 & \nabla q - \nabla \cdot \hat{\boldsymbol{\sigma}}(\mathbf{v}_s) = -(1-\phi_f) \rho_r \mathbf{g} \\
 & \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0
 \end{aligned}$$

List SI units

q = pascal.

$\mathbf{v}$ = m/s

Non dimensionalization choice

- $\tilde{\mathbf{v}} = \frac{\tilde{\mathbf{v}}_r}{k_0 \rho_r g_y / \mu_f}$
- $\check{q} = \frac{q}{\rho_r g_y l_0}$
- $\check{x} = \frac{x}{l_0}$
- $l_0 = \left(\frac{k_0 \mu_s}{\mu_f} \right)^{1/2}$

The resulting non dimensionalized form is therefore

$$\begin{aligned}
 (1.2) \quad & \tilde{\mathbf{v}}_r + \phi_f^{1+\Theta} \nabla (\phi_f^{-1/2} \check{q}_f) = 0 \\
 & \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) = 0 \\
 & \nabla \check{q} - \nabla \cdot \hat{\boldsymbol{\sigma}}(\check{\mathbf{v}}_s) = -(1-\phi_f) \rho_r \hat{\mathbf{g}} \\
 & \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f} (\check{q}_f - \phi_f^{1/2} \check{q}) = 0
 \end{aligned}$$

scaled nondimensionalized weak form let $\check{\boldsymbol{\sigma}} = \mathcal{D}\mathbf{v} - \frac{1}{3} \nabla \cdot \mathbf{v} \mathbf{I}$

$$\begin{aligned}
 (1.3) \quad & (2\phi_s \check{\boldsymbol{\sigma}}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\check{q}_h, \nabla \cdot \Psi_s) = -(\phi_s \check{\mathbf{f}}, \Psi_s) \\
 & (\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s} \check{q}_h, \omega \right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_{f,h}, \omega \right) = 0 \\
 & (\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r)) = 0 \\
 & (\hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s} \check{q}_{f,h}, \omega_f \right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \check{q}_h, \omega_f \right) = 0
 \end{aligned}$$

1.1. Variational form and discretization.

1.1.1. Stokes and variational formulation.

$$(1.4) \quad \begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

weak form

$$(1.5) \quad \begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_{\Omega} - (p, \nabla \cdot \mathbf{v})_{\Omega} &= (\mathbf{f}, \mathbf{v})_{\Omega} + \langle \boldsymbol{\tau}^T \nabla \mathbf{u} \mathbf{v}, \mathbf{v} \cdot \boldsymbol{\tau} \rangle_{\Gamma_i} \\ &+ \langle \mathbf{v}^T \nabla \mathbf{u} \mathbf{v} - p_0, \mathbf{v} \cdot \mathbf{v} \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

let stress $\mathbf{s} \cdot \boldsymbol{\tau} = \boldsymbol{\tau}^T \nabla \mathbf{u} \mathbf{v}$ and $\mathbf{s} \cdot \mathbf{v} = \mathbf{v}^T \nabla \mathbf{u} \mathbf{v} - p_0$, we have traction boundary condition term defined on Γ_i which is $\langle \mathbf{s}, \mathbf{v} \rangle_{\Gamma_i}$. Traditionally a test space $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0 \text{ on } \Gamma_u\}$ is employed here.

$$(1.6) \quad \begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

1.2. Darcy.

$$(1.7) \quad \begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

weak form

$$(1.8) \quad \begin{aligned} (\mathbf{u}, \mathbf{v})_{\Omega} - (p, \nabla \cdot \mathbf{v})_{\Omega} &= (\mathbf{g}, \mathbf{v})_{\Omega} - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

Test space $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

1.2.1. A degenerate Darcy test problem.

$$(1.9) \quad \begin{aligned} \mathbf{v} + \phi \nabla (\hat{\phi}^{-1/2} p) &= \mathbf{f} \\ \hat{\phi}^{-1/2} \nabla \cdot (\phi \mathbf{v}) &= g \end{aligned}$$

We manufacture a solution with degenerate porosity in the target physical domain $x, y \in [-1, 1]$.

$$(1.10) \quad \phi = \begin{cases} \cos^m(n(x^2 + y^2)) & x^2 + y^2 < \pi/2n \\ 0 & \text{otherwise} \end{cases}$$

where m is an even number, and $\pi/2n < 1$ another choice of test case is shown as follows:

$$(1.11) \quad \phi = \begin{cases} 0 & x \leq 0 \\ \phi_+ x^4 & x > 0 \end{cases}$$

Define an arbitrary velocity and pressure

$$(1.12) \quad \begin{aligned} \mathbf{v} &= \left[\frac{x}{5}, -y \right]^T \\ p &= \hat{\phi}^{1/2} \end{aligned}$$

The source term is calculated as follows:

$$(1.13) \quad f = \mathbf{v}$$

and the divergence term is:

$$(1.14) \quad \begin{aligned} g &= \phi^{-1/2} \phi_+ \nabla \cdot \left[\frac{x^5}{5}, -x^4 y \right]^T \\ &= \phi^{-1/2} \phi_+ (x^4 - x^4) = 0 \end{aligned}$$

a divergence free test case

1.2.2. Modified BR element. The Bernardi-Raugel element has been researched extensively in previous literature [?]. The error estimate is well known established by[?].

$$(1.15) \quad \|\mathbf{u} - \mathbf{u}\|_{1,\Omega} + \|p - p_h\|_{0,\Omega} \leq Ch^m (\|\mathbf{u}\|_{m+1,\Omega} + \|p\|_{m,\Omega})$$

We state that our new function space satisfies the following hypotheses such that the error estimate holds:

- For any q in $H^m(\Omega) \cap L_0^2(\Omega)$, one has

$$\inf_{q_h \in \mathbf{M}_h} \|q - q_h\|_{0,\Omega} \leq Ch^m \|q\|_{m,\Omega}$$

- there exists a linear operator Π_h from $H^{m+1}(\Omega)^d \cap H_0^1(\Omega)^d$ into X_h such that

$$\forall \mathbf{v} \in H^{m+1}(\Omega)^d \cap H_0^1(\Omega)^d, \begin{cases} \forall q_h \in M_h, (q_h, \operatorname{div}(\mathbf{v} - \Pi_h \mathbf{v})) = 0, \\ \|\mathbf{v} - \Pi_h \mathbf{v}\| \leq Ch^m \|\mathbf{v}\|_{m+1,\Omega} \end{cases}$$

- for each q_h in M_h , there exists a function $\mathbf{v}_h$ in X_h such that

$$(\operatorname{div} \mathbf{v}_h, q_h) \geq \beta \|q_h\|_{0,\Omega} \|\mathbf{v}_h\|_{1,\Omega}$$

, where $\beta > 0$ is a constant independent of h .

We introduce a family of quadrilateral mesh of domain Ω , bounded by h .

2. Linear system and Uzawa type algorithm. Boundary conditions are already included in the linear system formulation. Original linear system:

$$(2.1) \quad \begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}$$

This is a double saddle point system. Symmetrically permute original linear system and form an enlarged saddle point system.

$$(2.2) \quad \begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}$$

(2.3)

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}$$

In this newly formed system, the $\mathbf{A}$ matrix maintains as a positive definite matrix. we solve this typical saddle point system with uzawa iteration. Uzawa algorithm has been discussed extensively at [?], [?].

Consider the linear system of equation:

$$(2.4) \quad \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}$$

Algorithm 2.1 Preconditioned Inexact Uzawa Iteration

- 1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
 - 2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
 - 3: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \mathbf{Q}^{-1}(\mathbf{B}^T \mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$
-

In the previous literature [?], $\mathbf{Q}$ is approximated with $\tau\mathbf{I}$, which works well with the stokes part. However, it converges very slow with the same approximation when applied to the Darcy part. We need a better preconditioner approximating inverse of the corresponding schur complement. We use MINRES algorithm [?] due to schur complement being symmetric and indefinite.

Algorithm 2.2 Preconditioned Inexact Uzawa Iteration

- 1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
 - 2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
 - 3: Obtain $\mathbf{r} = \mathbf{B}^T \mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
 - 4: Obtain $\tilde{\mathbf{y}} = \operatorname{argmin}_{\mathbf{y} \in V} \|(\mathbf{B}^T \mathbf{A}^{-1} \mathbf{B} - \mathbf{C})^{-1} \tilde{\mathbf{y}} - \mathbf{r}\|$
 - 5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$
-

Based on the above algorithm, we proposed a new block preconditioner for our coupled linear system.

$$(2.5) \quad \mathbf{Q}^{-1} = \begin{bmatrix} \mathbf{S}_s^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_d^{-1} \end{bmatrix}$$

where the schur complement matrices are approximated with MINRES method.

2.1. Convergence and stability analysis. Uzawa iteration is in fact a fixed point iteration procedure. Usually, uzawa iteration convergences slowly.

2.1.1. Anderson acceleration. Accelerating uzawa iteration has been discussed in the previously literatures. Anderson acceleration (Anderson mixing) [?]

3. Test cases. We test the linear system a little bit different from the linear system above.

3.1. Constant porosity and balanced pressure (divergence free velocity field).

3.2. Const porosity and unbalanced pressure.

$$(3.1) \quad \check{q} = 0 \quad \check{q}_f = 2(x+y) \quad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \quad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$$

$$(3.2) \quad \nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f} (x+y)$$

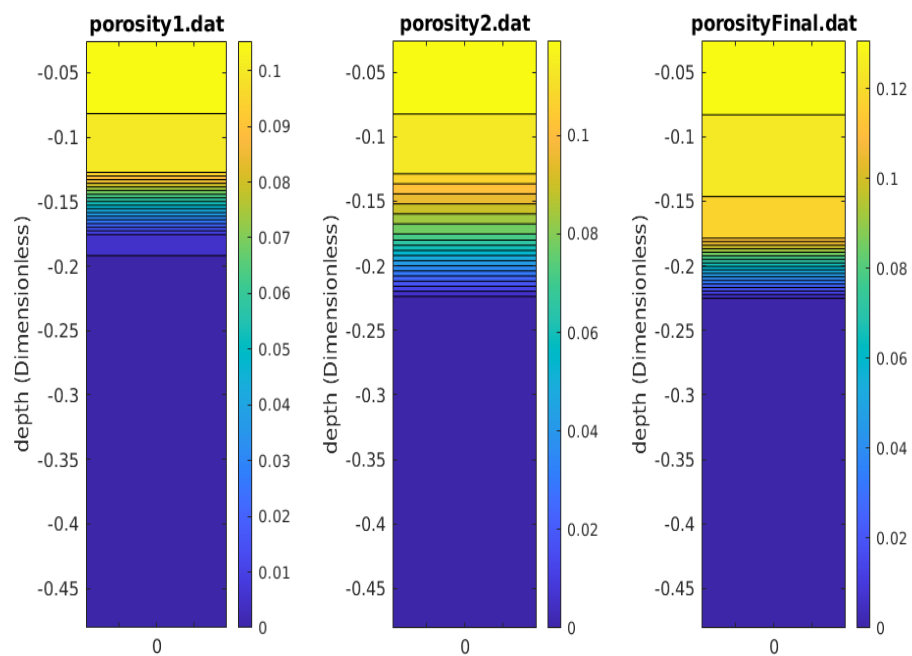
$$(3.3) \quad \mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f} \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \quad (\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I} = \frac{2\phi_f^{1/2}}{1-\phi_f} \begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix}$$

$$(3.4) \quad \begin{aligned} \check{\mathbf{v}}_r + \phi_f^{1/2} \nabla \check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2} \begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \phi_f^{1/2} \nabla \cdot (\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f} \check{q}_f &= \frac{-2}{1-\phi_f} (x+y) + \frac{1}{1-\phi_f} 2(x+y) = 0 \\ \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f} \check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f} (x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f} (x+y) = 0 \\ -2(1-\phi_f) \nabla \cdot \boldsymbol{\sigma}(\check{\mathbf{v}}_s) &= -2(1-\phi_f) \nabla \cdot (\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3} (\nabla \cdot \check{\mathbf{v}}_s) \mathbf{I}) = -\frac{8\phi_f^{1/2}}{3} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned}$$

4. Retrieve of physical variables. Eventually, we need to retrieve actual physicia values from nondimensionalized scaled variables. We need original velocity and pressure.

As stated above, the nondimensionalized scaled pressure potential $\check{q} = \frac{q}{\rho_r g_y l_0}$, hence $q = \rho_r g_y l_0 \check{q}$. The pressure potentials eliminated the effect of gravity, where $q_f = p_f - \rho_f g z$ and $q_s = p_s - \rho_f g z$. We can retrieve of original velocity with the similar manner $\tilde{\mathbf{v}} = \phi_f^{-1-\Theta} \mathbf{v}_r$.

Porosity plots



Initial velocity distribution

